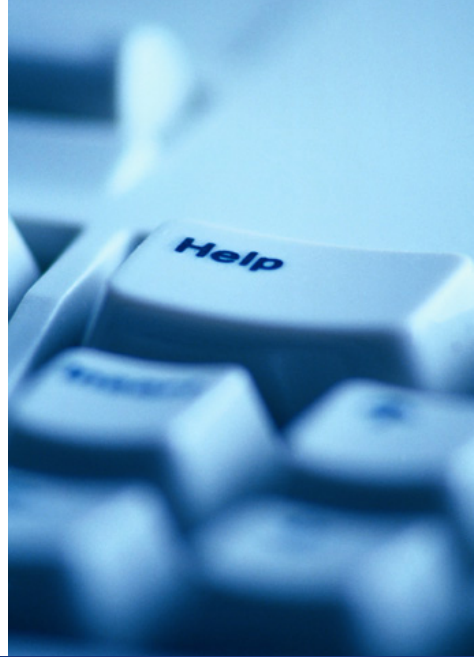




操作系统原理

Operating Systems Principles

陈鹏飞
计算机学院



中山大學
SUN YAT-SEN UNIVERSITY

计算机学院（软件学院）

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING



第十三讲 — 文件系统





目标

- 文件系统的功能；
- 文件系统接口；
- 文件系统的设计权衡，访问方法、共享、加锁以及目录结构；
- 文件系统保护；



文件

- ❖ 操作系统对存储设备的物理属性加以抽象，从而定义逻辑存储单位，即文件，文件由操作系统映射到物理设备上。
- ❖ 用户创建的数据集合；
- ❖ 对于用户来说，文件系统是操作系统中最重要的部分之一，主要特点如下：

Long-term existence

- files are stored on disk or other secondary storage and do not disappear when a user logs off

Sharable between processes

- files have names and can have associated access permissions that permit controlled sharing

Structure

- files can be organized into hierarchical or more complex structure to reflect the relationships among files



文件概念

- ❖ 连续的逻辑地址空间
- ❖ 类型:
 - 数据
 - Numeric
 - Character
 - Binary
 - 程序
- ❖ 内容由文件的创建者定义
 - 很多类型
 - **text file,**
 - **source file,**
 - **executable file**



文件属性

- ❖ 名称——仅以人类可读形式保存的信息
- ❖ 标识符——唯一标识（编号）用于标识文件系统中的文件
- ❖ 类型——支持不同类型的系统所需
- ❖ 位置——指向设备上文件位置的指针
- ❖ 大小——当前文件大小
- ❖ 保护——控制谁可以读、写、执行
- ❖ 时间、日期和用户标识——用于保护、安全和使用监控的数据
- ❖ 有关文件的信息保存在磁盘上维护的目录结构中
- ❖ 许多变体，包括扩展文件属性，如文件校验和
- ❖ 目录结构中保存的信息



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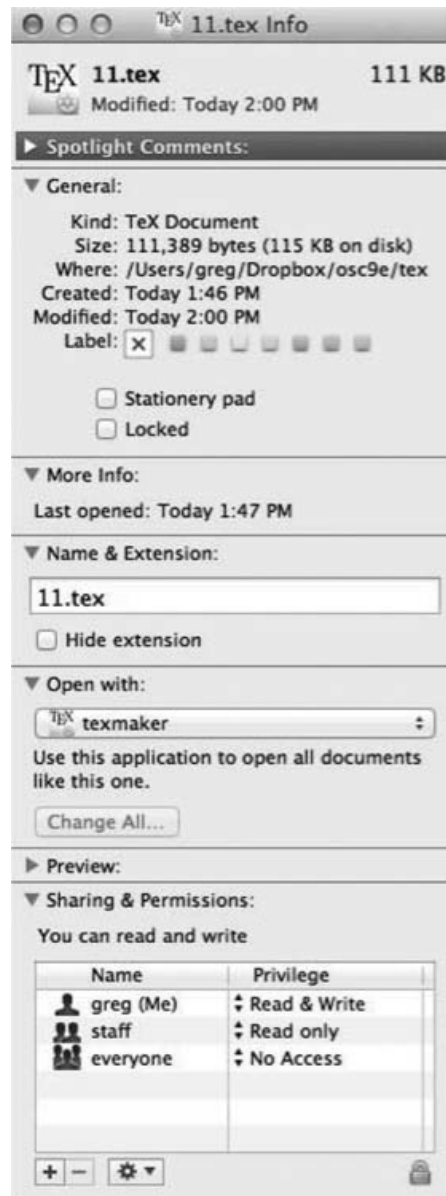
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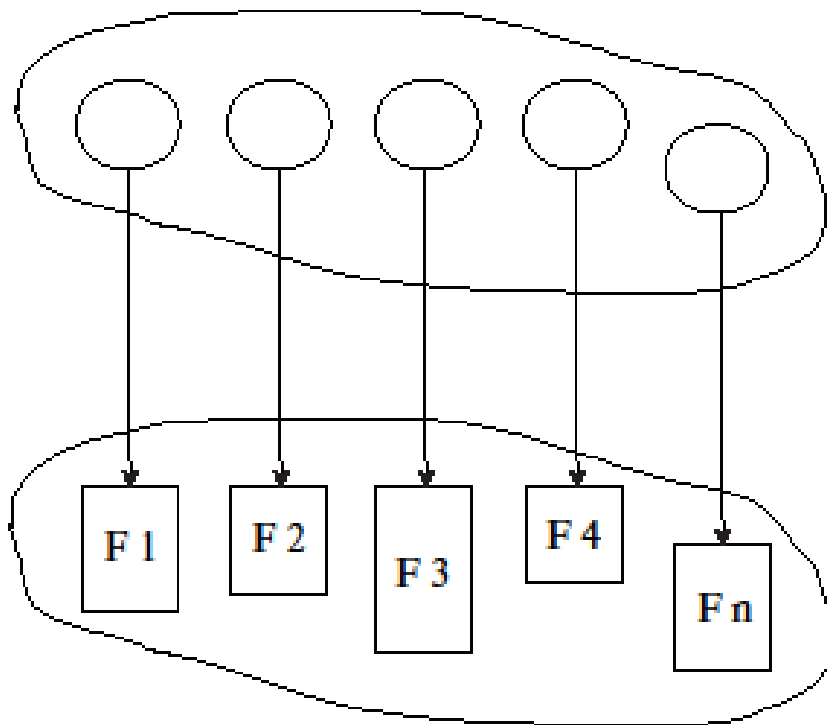
Mac OS上的文件属性





文件目录结构

- ❖ A collection of nodes containing information about all files



- ❖ Both the directory structure and the files reside on disk



文件操作

- ❖ Create
- ❖ Write – at write pointer location
- ❖ Read – at read pointer location
- ❖ Reposition within file - seek
- ❖ Delete
- ❖ Truncate (截断)
- ❖ *Open* (F_i) – search the directory structure on disk for entry F_i , and **move the content of entry to memory**
- ❖ *Close* (F_i) – move the content of entry F_i in memory to directory structure on disk

这6个基本操作组所需文件操作的最小集合



文件打开 (Open)

❖ Several pieces of data are needed to manage open files:

- **Open-file table:** tracks open files (进程打开表、系统打开表)
- **File pointer:** pointer to last read/write location, per process that has the file open
- **File-open count:** counter of number of times a file is open – to allow removal of data from open-file table when last processes closes it;
- **Disk location of the file:** cache of data access information
- **Access rights (访问权限) :** per-process access mode information



文件锁

- ❖ **Provided by some operating systems and file systems**
 - Similar to reader-writer locks
 - **Shared lock** (共享锁) similar to reader lock – several processes can acquire concurrently
 - **Exclusive lock** (互斥锁) similar to writer lock
- ❖ **Mediates access to a file** (控制对文件的访问)
- ❖ **Mandatory** (强制性的) **or advisory** (建议性的) :
 - **Mandatory** – access is denied depending on locks held and requested (Windows)
 - **Advisory** – processes can find status of locks and decide what to do (Unix)



文件锁案例——Java

```
import java.io.*;
import java.nio.channels.*;
public class LockingExample {
    public static final boolean EXCLUSIVE = false;
    public static final boolean SHARED = true;
    public static void main(String arsg[]) throws IOException
    {
        FileLock sharedLock = null;
        FileLock exclusiveLock = null;
        try {
            RandomAccessFile raf = new
RandomAccessFile("file.txt", "rw");
            // get the channel for the file
            FileChannel ch = raf.getChannel();
            // this locks the first half of the file - exclusive
            exclusiveLock = ch.lock(0, raf.length()/2,
EXCLUSIVE);
            /** Now modify the data . . . */
            // release the lock
            exclusiveLock.release();
```



文件锁案例——Java

```
// this locks the second half of the file - shared
sharedLock = ch.lock(raf.length()/2+1,
raf.length(), SHARED);
/** Now read the data . . . */
// release the lock
sharedLock.release();
} catch (java.io.IOException ioe) {
    System.err.println(ioe);
}finally {
    if (exclusiveLock != null)
        exclusiveLock.release();
    if (sharedLock != null)
        sharedLock.release();
}
}
```



文件类型

file type	usual extension	function
executable	exe, com, bin or none	ready-to-run machine-language program
object	obj, o	compiled, machine language, not linked
source code	c, cc, java, pas, asm, a	source code in various languages
batch	bat, sh	commands to the command interpreter
text	txt, doc	textual data, documents
word processor	wp, tex, rtf, doc	various word-processor formats
library	lib, a, so, dll	libraries of routines for programmers
print or view	ps, pdf, jpg	ASCII or binary file in a format for printing or viewing
archive	arc, zip, tar	related files grouped into one file, sometimes compressed, for archiving or storage
multimedia	mpeg, mov, rm, mp3, avi	binary file containing audio or A/V information

Windows通过扩展名识别文件类型，Unix系统通过文件开始部分的magic数字来识别类型



文件结构

- ❖ None - sequence of words, bytes
- ❖ Simple record structure
 - Lines
 - Fixed length
 - Variable length
- ❖ Complex Structures
 - Formatted document
 - Relocatable load file
- ❖ **Can simulate last two with first method by inserting appropriate control characters**
- ❖ Who decides:
 - Operating system
 - Program



文件访问方法

- 顺序访问
- 直接访问
- 其他访问方法

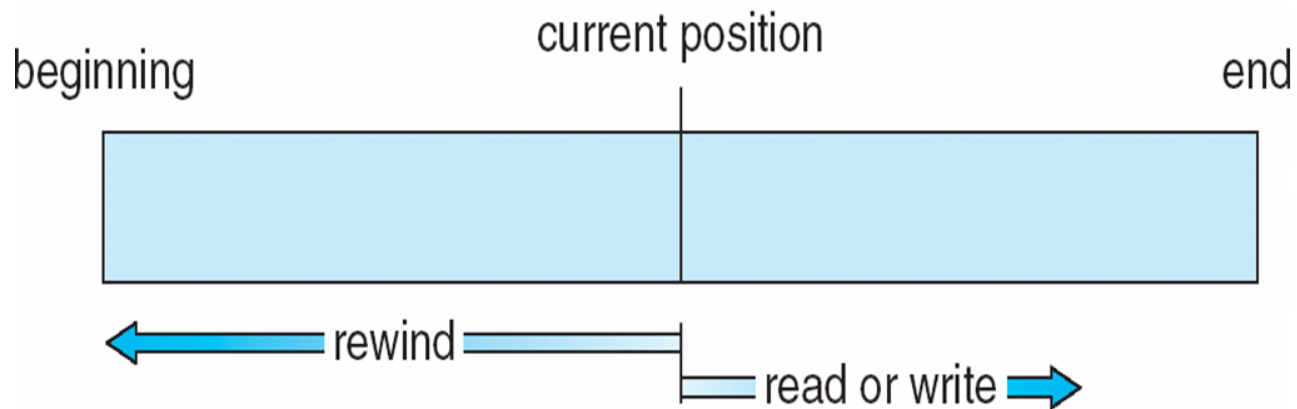


顺序访问

❖ Operations

- read next
- write next
- Reset

❖ Figure





直接访问

❖ **Operations** 适用于由固定长度的逻辑记录组成的文件；

- **read n**
- **write n**
- **position to n**
 - **read next**
 - **write next**
 - **rewrite n**

n = relative block number

❖ **Relative block numbers (相对块号) allow OS to decide where file should be placed**



直接访问模拟顺序访问

sequential access	implementation for direct access
<i>reset</i>	$cp = 0;$
<i>read next</i>	$read\ cp;$ $cp = cp + 1;$
<i>write next</i>	$write\ cp;$ $cp = cp + 1;$



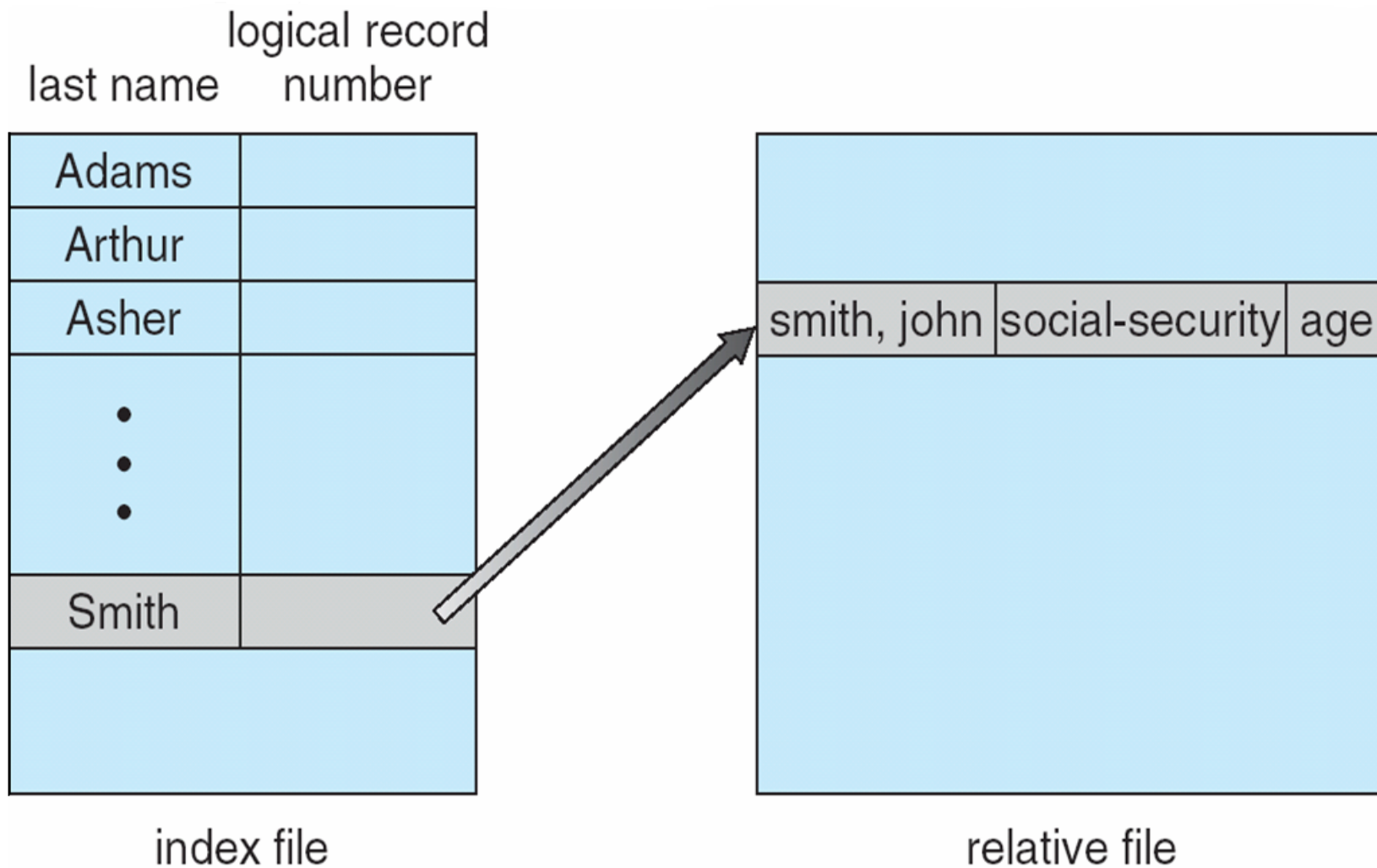
其他访问方法

- 可以在基础方法之上构建其他访问方法
- 通常需要为文件创建索引
- 将索引保存在内存中，以便快速确定要操作的数据位置（例如通用产品代码（UPC 码）及其相关商品数据记录）
- 如果索引过大，则创建内存索引，即磁盘索引的索引
- IBM 的索引顺序访问方法（ISAM）
 - 小型主索引指向二级索引的磁盘块
 - 文件按定义的键保持排序状态
 - 所有这些均由操作系统完成

通过记录的键至多两次的直接访问就可以定位记录
- 作为另一个示例，VMS 操作系统提供了索引文件和相关文件（见下一张幻灯片）



索引文件和相关文件的例子



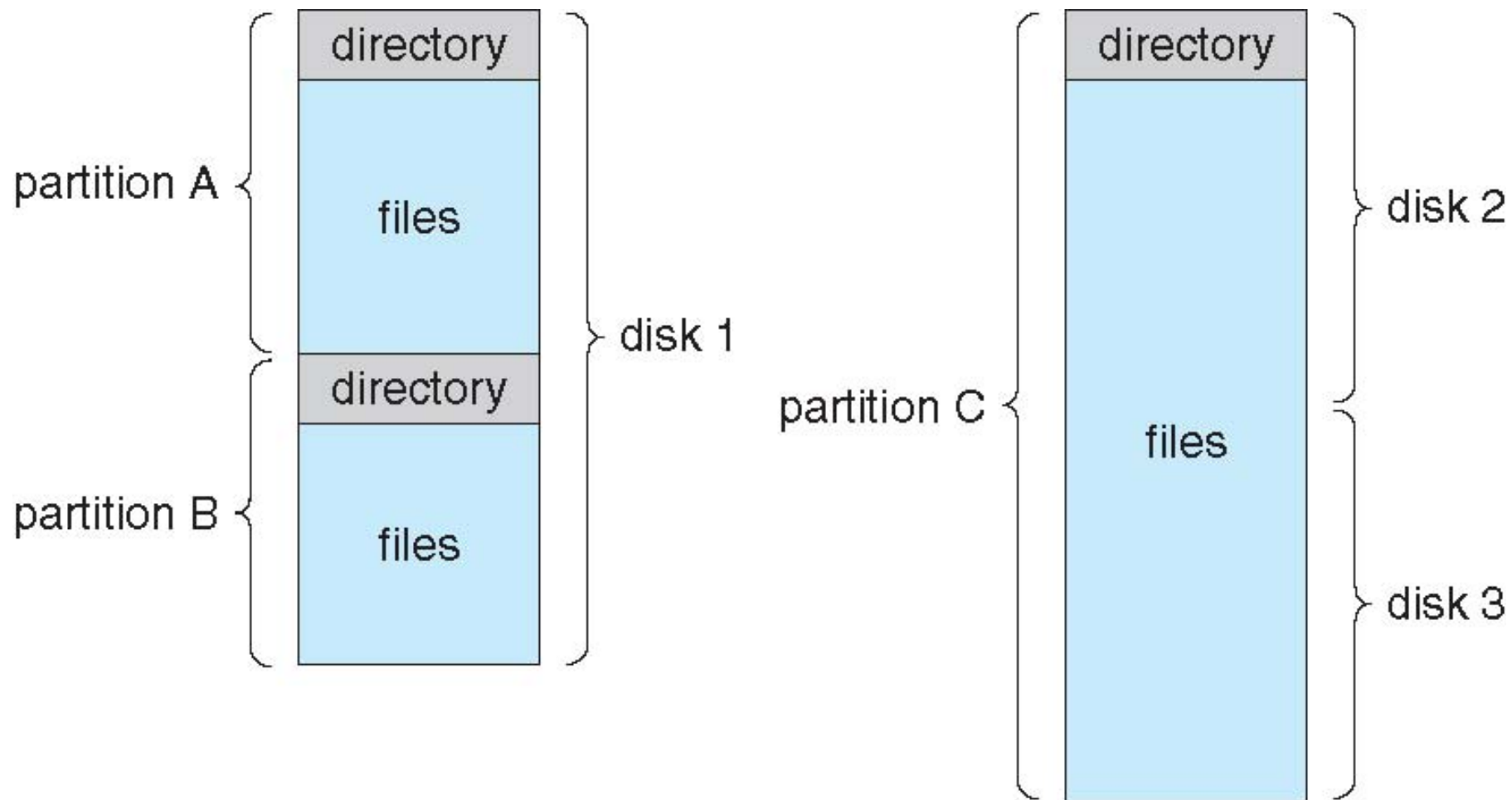


磁盘结构

- ❖ 磁盘可以细分为多个分区
- ❖ 磁盘或分区可以进行**RAID**保护以防止故障
- ❖ 磁盘或分区可以在没有文件系统的情况下直接使用，也可以使用文件系统进行格式化
- ❖ 分区也称为迷你磁盘、分片
- ❖ 包含文件系统的实体称为**卷**
- ❖ 每个包含文件系统的卷还会在设备目录或卷目录中跟踪该文件系统的信息
- ❖ 除了通用文件系统外，还有许多专用文件系统，通常都在同一操作系统或计算机中



存储结构





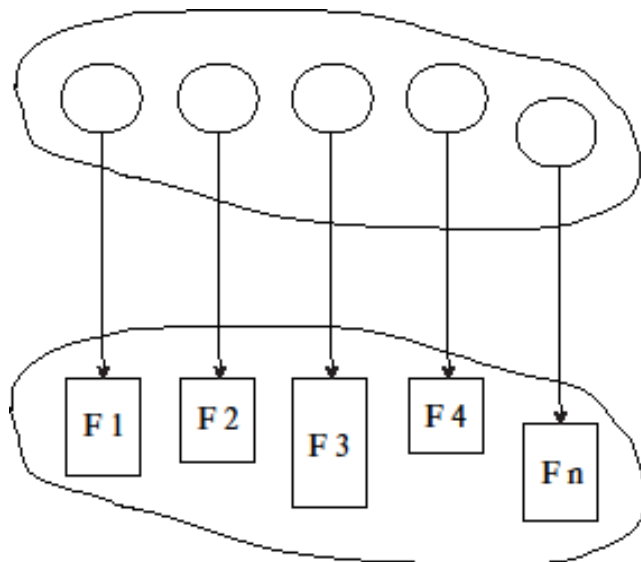
文件系统的类型

- ❖ We mostly talk of general-purpose file systems
- ❖ But systems frequently have many file systems, some general- and some special- purpose
- ❖ Consider Solaris has
 - tmpfs – memory-based volatile FS for fast, temporary I/O
 - objfs – interface into kernel memory to get kernel symbols for debugging
 - ctfs – contract file system for managing daemons
 - lofs – loopback file system allows one FS to be accessed in place of another
 - procfs – kernel interface to process structures
 - ufs, zfs – general purpose file systems



目录结构

- ❖ A collection of nodes containing information about all files
- ❖ 符号表，可以将文件名称转成目录条目



- ❖ Both the directory structure and the files reside on disk



目录结构

- ❖ Search for a file
- ❖ Create a file
- ❖ Delete a file
- ❖ List a directory
- ❖ Rename a file
- ❖ Traverse the file system



目录结构

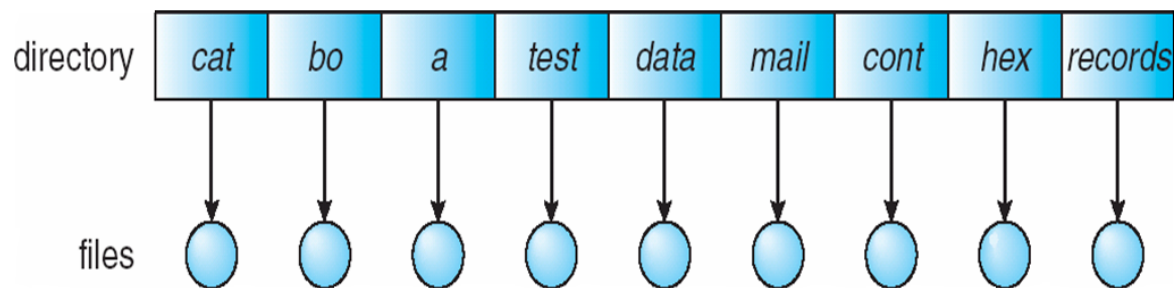
The directory is organized logically to obtain

- ❖ **Efficiency – locating a file quickly**
- ❖ **Naming – convenient to users**
 - Two users can have same name for different files
 - The same file can have several different names
- ❖ **Grouping – logical grouping of files by properties, (e.g., all Java programs, all games, ...)**



单级目录

❖ A single directory for all users



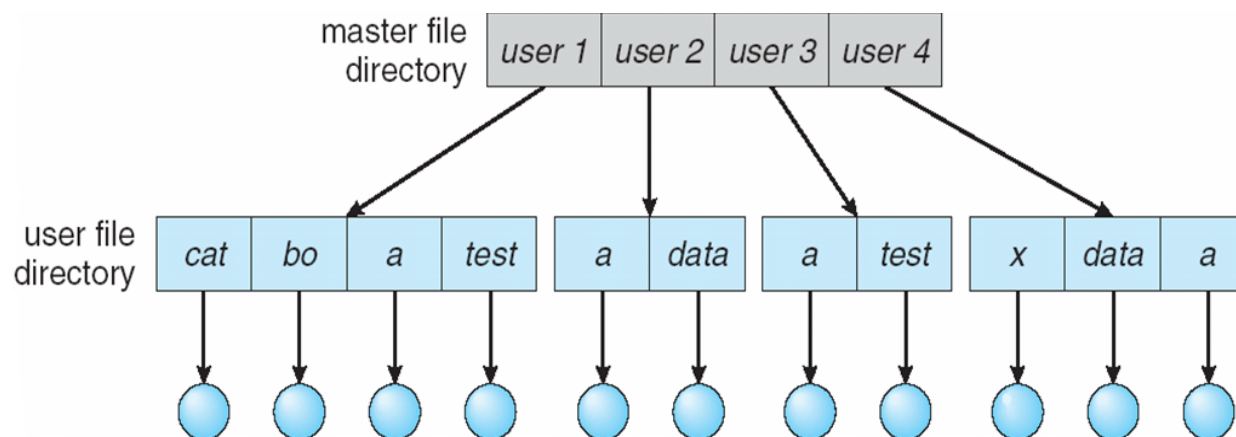
❖ Naming problem

❖ Grouping problem



两级目录结构

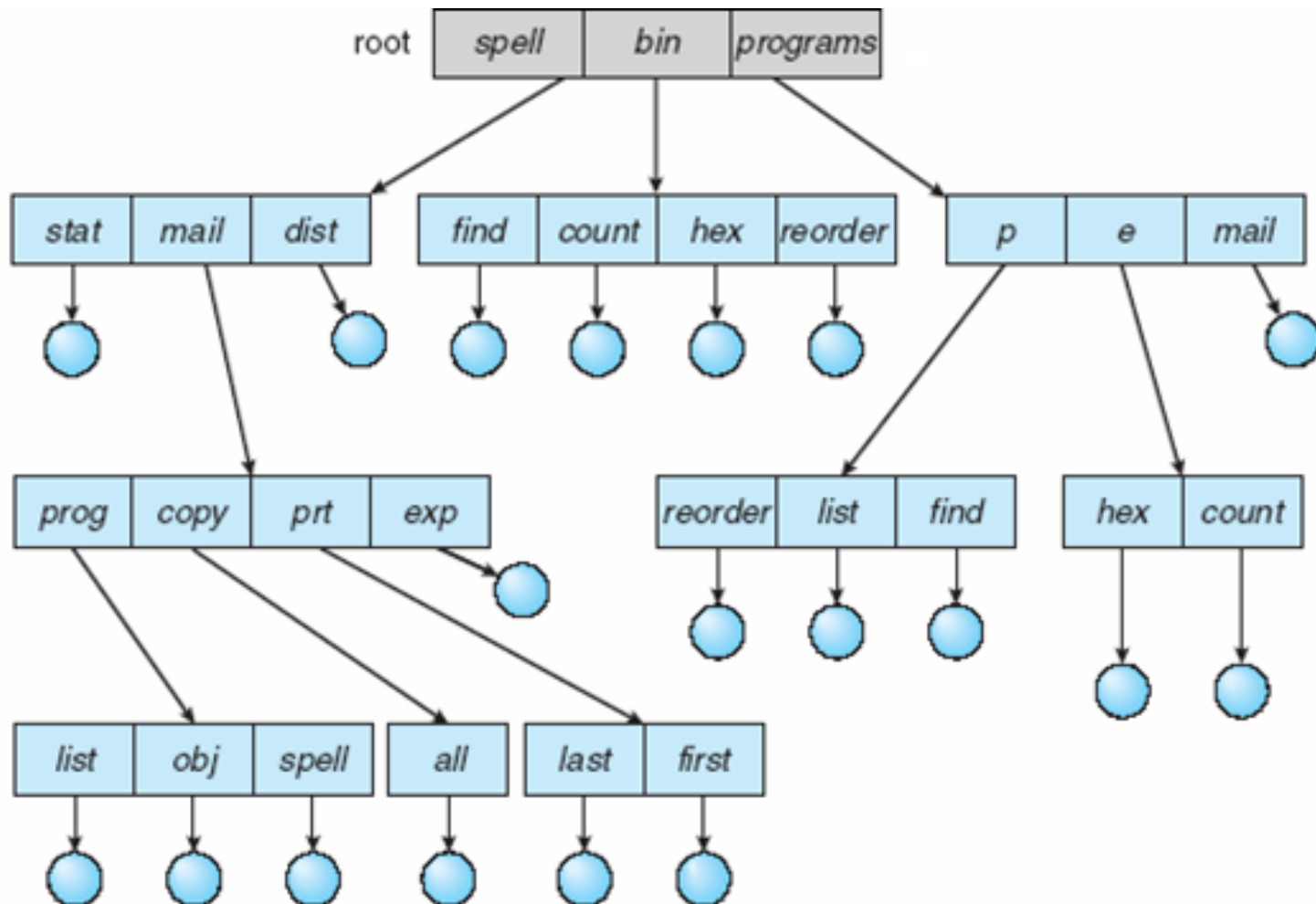
❖ Separate directory for each user



- Path name
- Can have the same file name for different user
- Efficient searching



树形目录







无环图目录

- ❖ Two different names (aliasing)
- ❖ If *dict* deletes w/*list* $\Rightarrow$ dangling pointer

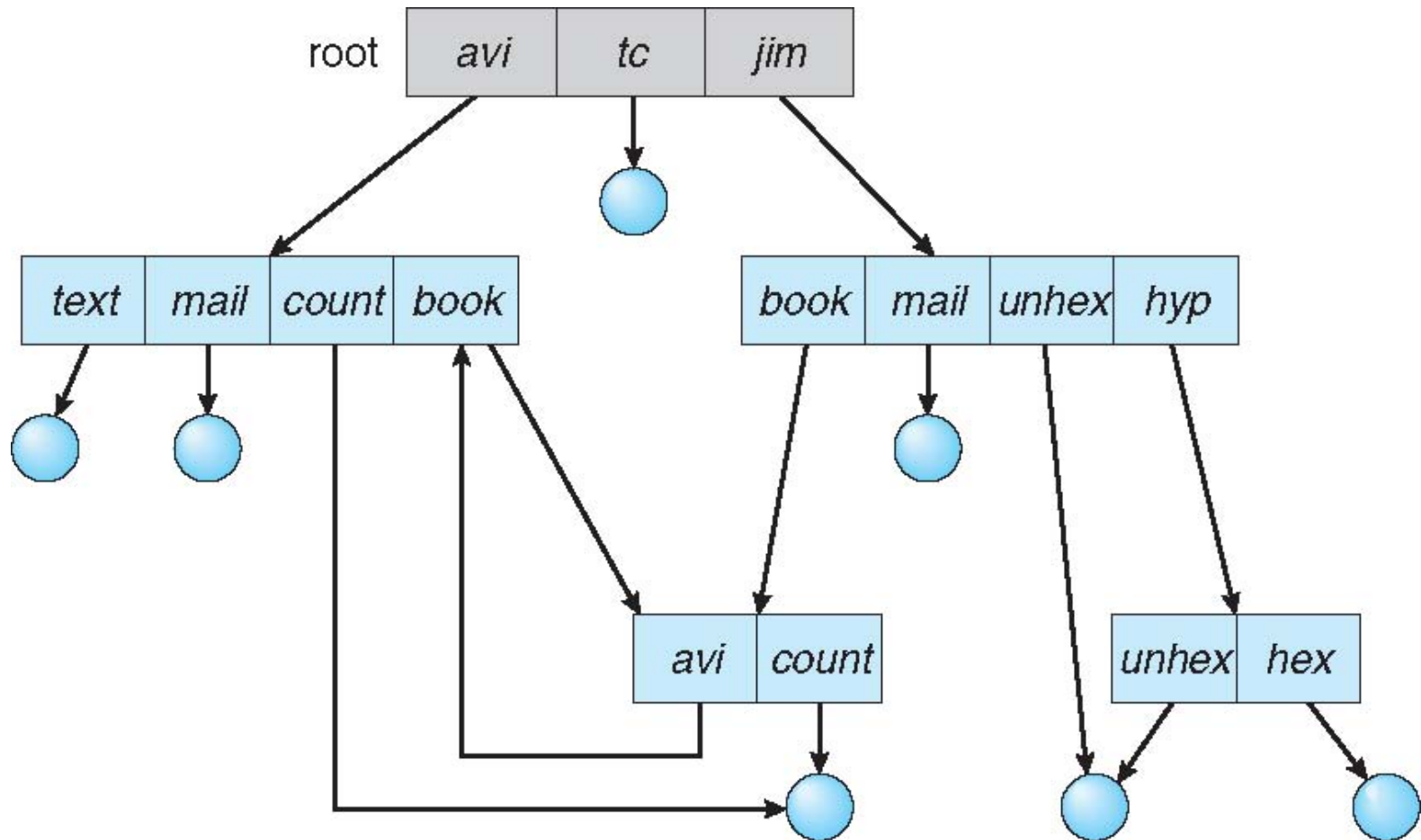
Solutions:

- Backpointers, so we can delete all pointers.
 - Variable size records a problem
- Backpointers using a daisy chain organization
- Entry-hold-count solution
- ❖ New directory entry type
 - **Link** – another name (pointer) to an existing file
 - **Resolve the link** – follow pointer to locate the file

软链接、硬链接



通用图目录





通用图目录

❖ How do we guarantee no cycles?

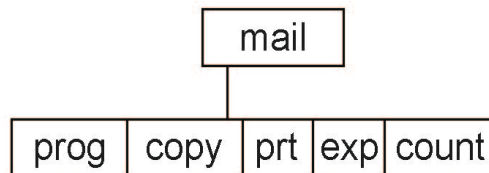
- Allow only links to files not subdirectories
- **Garbage collection**
- Every time a new link is added use a cycle detection algorithm to determine whether it is OK



当前目录

- ❖ Can designate one of the directories as the current (working) directory
 - `cd /spell/mail/prog`
 - `type list`
- ❖ Creating and deleting a file is done in current directory
- ❖ Example of creating a new file
 - If in current directory is **/mail**
 - The command
`mkdir <dir-name>`

- Results in:



- Deleting “mail” $\Rightarrow$ deleting the entire subtree rooted by “mail”



文件系统保护

❖ File owner/creator should be able to control:

- What can be done
- By whom

❖ Types of access

- Read
- Write
- Execute
- Append
- Delete
- List



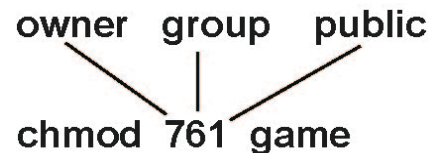
Linux文件访问

- ❖ Mode of access: read, write, execute
- ❖ Three classes of users on Unix / Linux

			RWX
a) owner access	7	⇒	1 1 1 RWX
b) group access	6	⇒	1 1 0 RWX
c) public access	1	⇒	0 0 1

- ❖ Ask manager to create a group (unique name), say G, and add some users to the group.
- ❖ For a file (say *game*) or subdirectory, define an appropriate access.

chgrp G game



- ❖ Attach a group to a file 37



Unix文件目录

-rw-rw-r--	1 pbg	staff	31200	Sep 3 08:30	intro.ps
drwx-----	5 pbg	staff	512	Jul 8 09:33	private/
drwxrwxr-x	2 pbg	staff	512	Jul 8 09:35	doc/
drwxrwx---	2 pbg	student	512	Aug 3 14:13	student-proj/
-rw-r--r--	1 pbg	staff	9423	Feb 24 2003	program.c
-rwxr-xr-x	1 pbg	staff	20471	Feb 24 2003	program
drwx--x--x	4 pbg	faculty	512	Jul 31 10:31	lib/
drwx-----	3 pbg	staff	1024	Aug 29 06:52	mail/
drwxrwxrwx	3 pbg	staff	512	Jul 8 09:35	test/



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谢谢